1 Data Structures for Disjoint Sets

Disjoint sets (Union-Find)

Each set identified by a representative.

Operations

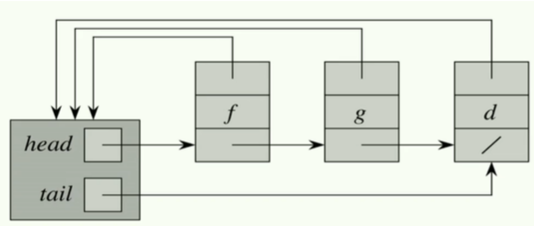
- MAKE-SET(x): $S_i = \{x\}$, add to $\mathcal{S}$

Let $x \in S_x$, $y \in S_y$

- UNION(x, y): $\mathcal{S} = \mathcal{S} \setminus \{S_x, S_y\} \cup \{S_x \cup S_y\}$
- FIND(x): return representative of x in S_x

```
CONNECTED-COMPONENTS(G)
    for each vertex v in G.V
        MAKE-SET(v)
    for each edge (u, v) in G.E
        if FIND(u) != FIND(v)
            UNION(u, v)
```

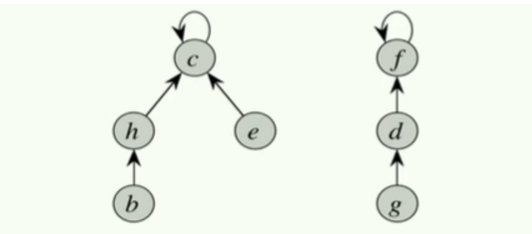
Linked list representation



Weighted-union heuristic

Always merge smaller into larger. m ops on n elements: $O(m + n \lg n)$. Each element's representative updated $\leq \lg n$ times (each update at least doubles set size).

Tree representation

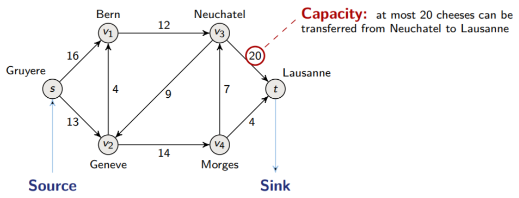


Runtime with both heuristics

Union by rank: smaller-rank root $\rightarrow$ child of larger-rank root (rank $\approx$ height upper bound).
Path compression: all nodes on FIND path become direct children of root.
Union by rank + path compression: $O(m \cdot \alpha(n))$ for m operations on n elements. $\alpha(n) \leq 5$ for any practical n. Bound is tight.

2 Flow network

Definitions



Formally

- Source: $s \in V$, sink: $t \in V$
- Directed graph (DG): $G = (V, E)$
- Each (u, v) has a capacity $c(u, v) \geq 0$
- No antiparallel: if $(u, v) \in E$, then $(v, u) \notin E$

Capacity constraint

$\forall u, v \in V$, $0 \leq f(u, v) \leq c(u, v)$

Flow conservation

$\forall u \in V \setminus \{s, t\}$, $\sum_{v \in V} f(u, v) = \sum_{v \in V} f(v, u)$

Cut

Partition of V into two disjoint sets S and T such that $s \in S$ and $t \in T$. $c(S, T) = \sum_{u \in S, v \in T} c(u, v)$.

Maximum flow problem - Ford-Fulkerson method

Residual network $G_f = (V, E_f)$

$c_f(u, v)$: additional flow possible (forward edge) or removable (backward edge).

```
FORD-FULKERSON-METHOD(G, s, t)
    initialize flow f to 0
    while exists an augmenting path p in G_f
        b = bottleneck(p)
        augment flow along p by b
    return f
```

Min cut problem

Min cut

It is a cut where the capacity crossing it is the minimum among all cuts.

Net flow accross cut

$f(S, T) = \sum_{u \in S, v \in T} f(u, v) - \sum_{u \in S, v \in T} f(v, u)$

Max-flow min-cut theorem ($\Leftrightarrow$)

- f is the maximum flow
- G_f has no augmenting path
- $|f| = c(S, T)$ for a minimum cut (S, T)

Applications of max flow

Bipartite matching as flow

$G = (V, E)$, $V = L \cup R$. Connect $s \rightarrow \forall u \in L$ and $\forall v \in R \rightarrow t$, all capacities 1. Run Ford-Fulkerson.

Edge-disjoint paths

Max # edge-disjoint paths from s to t = min edges to remove to disconnect s from t = max flow (all capacities 1).

3 Spanning Trees

Definition

Set T of edges that is acyclic and spanning (connects all vertices of G). **Goal of MST problem:** Find a spanning tree of min total weight.

Cut property (safe edge)

Given any cut, a minimum-weight crossing edge belongs to some MST.

Prim's algorithm

```
PRIM(G, w, r)    // G undirected ; w: E -> R weights
    Q = {}          // min-priority queue (min-heap)
    for each u in G.V:
        u.key = inf
        u.pi = NIL              // u.pi = parent in T
        INSERT(Q, u)
    DECREASE-KEY(Q, r, 0)                // r.key = 0
    while Q != {}:
        u = EXTRACT-MIN(Q)
        for each v in G.Adj[u]:
            if v in Q and w(u,v) < v.key:
                v.pi = u
                DECREASE-KEY(Q, v, w(u,v))
```

Why does it work?

T is always a subtree of a MST.

Prim's runtime

$O(E \lg V)$ with binary heap. $O(V \lg V + E)$ with Fibonacci heap.

Kruskal's algorithm

```
KRUSKAL(G, w)
    A = {}
    for v in G.V: MAKE-SET(v)
    sort G.E by w non-decreasing
    for (u, v) in G.E (sorted):
        if FIND(u) != FIND(v)
            A += {(u,v)}; UNION(u, v)
    return A
```

Why does it work?

T is always a sub-forest of a MST.

Kruskal's runtime

$O(E \lg E) = O(E \lg V)$ (sorting dominates).

4 Shortest Paths

Problem variants

- **Single-source:** from s to all vertices
- **Single-dest.:** reverse edges $\to$ single-source
- **Single-pair:** no better w.c. than single-source
- **All-pairs:** solve single-source for each vertex

Negative weights

Allowed if no negative-weight cycle is reachable from s. Dijkstra requires $w \geq 0$.

Optimal substructure

Any subpath of a shortest path is itself a shortest path (proof by contradiction).

```
INIT-SINGLE-SOURCE(G, s)
    for v in G.V: v.d = inf; v.pi = NIL
    s.d = 0

RELAX(u, v, w)
    if v.d > u.d + w(u, v)
        v.d = u.d + w(u, v); v.pi = u
```

Bellman-Ford

```
BELLMAN-FORD(G, w, s)
    INIT-SINGLE-SOURCE(G, s)
    for i = 1 to |G.V| - 1
        for (u, v) in G.E: RELAX(u, v, w)
    for (u, v) in G.E        // neg-cycle check
        if v.d > u.d + w(u, v): return FALSE
    return TRUE
```

Correctness

No neg-cycle reachable from $s \Rightarrow$ after $|V|-1$ iterations, $v.d = \delta(s,v)\ \forall v$. Shortest path uses $\leq |V|-1$ edges (can always remove cycle if non-negative weight).

Negative cycle detection

Neg-cycle reachable from $s \Leftrightarrow$ some ℓ-value changes in the n-th (extra) iteration.
Proof: If no change, $\forall(u,v) \in E : \ell(u) + w(u,v) \geq \ell(v)$. Summing over any cycle gives $\sum w(v_{i-1}, v_i) \geq 0$.

Runtime

$\Theta(E \cdot V)$. Good for distributed routing: each vertex repeatedly asks neighbors for best path.

Dijkstra's Algorithm

Non-negative weights only. Greedy (essentially weighted BFS, like Prim's).

```
DIJKSTRA(G, w, s)
    INIT-SINGLE-SOURCE(G, s)
    Q = G.V   // min-heap on d
    while Q != {}:
        u = EXTRACT-MIN(Q)
        for v in G.Adj[u]: RELAX(u, v, w)
```

Runtime

$O(E \lg V)$ with binary heap. $O(V \lg V + E)$ with Fibonacci heap.

5 Probabilistic Analysis

Motivation

Worst case rarely happens. Randomization helps avoid worst-case and adversarial inputs (e.g., random pivot in quicksort). Also necessary in cryptography.

Indicator Random Variables

Indicator RV

Given sample space and event A:

$$I\{A\} = \begin{cases} 1 & A \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

Lemma: $\mathbb{E}[I\{A\}] = \Pr\{A\}$ (since $\mathbb{E}[X_A] = 1 \cdot \Pr\{A\} + 0 \cdot \Pr\{\bar{A}\}$)

Linearity of expectation

$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$
Holds even when X and Y are dependent!

Hiring Problem

Hire candidate if taller than current best; n candidates, random arrival order.

- Worst case: n hires (candidates arrive sorted)
- Let $X_i = I\{\text{candidate } i \text{ hired}\}$, $X = \sum_{i=1}^{n} X_i$
- $\Pr\{\text{candidate } i \text{ hired}\} = \frac{1}{i}$ (is tallest among first i)
- By linearity: $\mathbb{E}[\#\text{hires}] = \sum_{i=1}^{n} \frac{1}{i} = H_n = \ln n + O(1)$

Expected number of hires is $O(\ln n)$ vs worst case $O(n)$.

6 Hash Tables

Hash function

$h : U \to \{0, \ldots, m-1\}$ maps key universe to m slots.
Collision: $h(k_1) = h(k_2)$ with $k_1 \neq k_2$.
Load factor: $\alpha = n/m$ (number of keys / slots).

Chaining (collision resolution)

Each slot: linked list of keys. Insert $O(1)$. Search/Delete $\Theta(1+\alpha)$ average. If $n = O(m)$ then $\alpha = O(1)$: all ops $O(1)$.

Division method

$h(k) = k \bmod m$. Avoid $m = 2^p$ (only uses low-order bits of k).

Multiplication method

$h(k) = \lfloor m \cdot (kA \bmod 1) \rfloor, \quad 0 < A < 1$.

Birthday paradox

n items into m slots: collision likely when $n \gtrsim \sqrt{2m}$.
$\Pr[\text{collision}] \geq \frac{1}{2}$ when $n \geq \sqrt{2m \ln 2} \approx 1.18\sqrt{m}$.

Birthday lemma

Prob. of *no* collision among q items drawn uniformly from $[n]$: $\leq e^{-q(q-1)/(2n)}$.

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Key definiti2ons

Linear map

$T: V \to W$ is linear if $T(\alpha x + \beta y) = \alpha T(x) + \beta T(y)$ for all $x, y \in V$, $\alpha, \beta \in \mathbb{R}$.

Rank–nullity

For $T: V \to W$ with $\dim V < \infty$:

$$\dim \ker T + \dim \operatorname{im} T \;=\; \dim V.$$

Formulas to remember

- **Quadratic:** $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- **Euler:** $e^{i\theta} = \cos\theta + i \sin\theta$
- **Geom. sum:** $\sum_{k=0}^{n} r^k = \dfrac{1 - r^{n+1}}{1 - r}$, $r \neq 1$
- **Taylor:** $f(x) = \sum_{k \geq 0} \dfrac{f^{(k)}(a)}{k!}(x - a)^k$

Pitfall

$\log(ab) = \log a + \log b$ *only* when $a, b > 0$.

Useful identities

$$\sin^2\theta + \cos^2\theta = 1$$
$$\cos(2\theta) = \cos^2\theta - \sin^2\theta$$
$$\sin(2\theta) = 2 \sin\theta \cos\theta$$

8 Topic 2 — Probability

Basics

- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$
- $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$
- Bayes: $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}$

Common distributions

Dist.	PMF/PDF	$\mathbb{E}[X]$
Bern(p)	$p^x(1-p)^{1-x}$	p
Bin(n, p)	$\binom{n}{x} p^x (1-p)^{n-x}$	np
Poiss(λ)	$e^{-\lambda}\lambda^x / x!$	λ
$\mathcal{N}(\mu, \sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$	μ

Worked

$X \sim \mathcal{N}(0, 1)$. Then $\mathbb{P}(|X| < 1) \approx 0.6827$.

9 Topic 3 — Algorithms

Complexity cheat

Op	Cost
Sort (comparison)	$\Theta(n \log n)$
Hash lookup (avg)	$\Theta(1)$
BFS / DFS	$\Theta(V + E)$
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Code snippet

```
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10 Quick reference

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